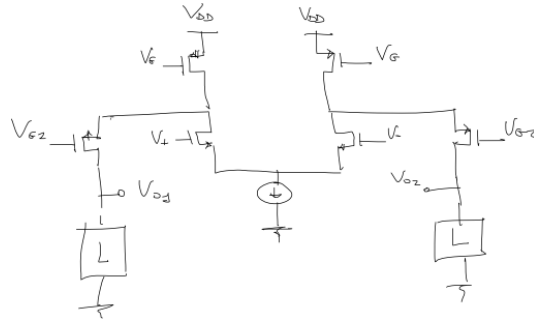


Lecture-20

→ Review of Lecture-19

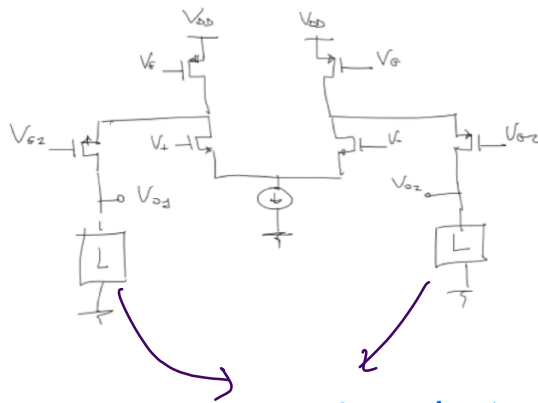
Folded Cascode
Diff. Output



Folded Cascode
Single-ended Output
?



Q How to
Draw Folded Cascode single ended output
Differential pair?



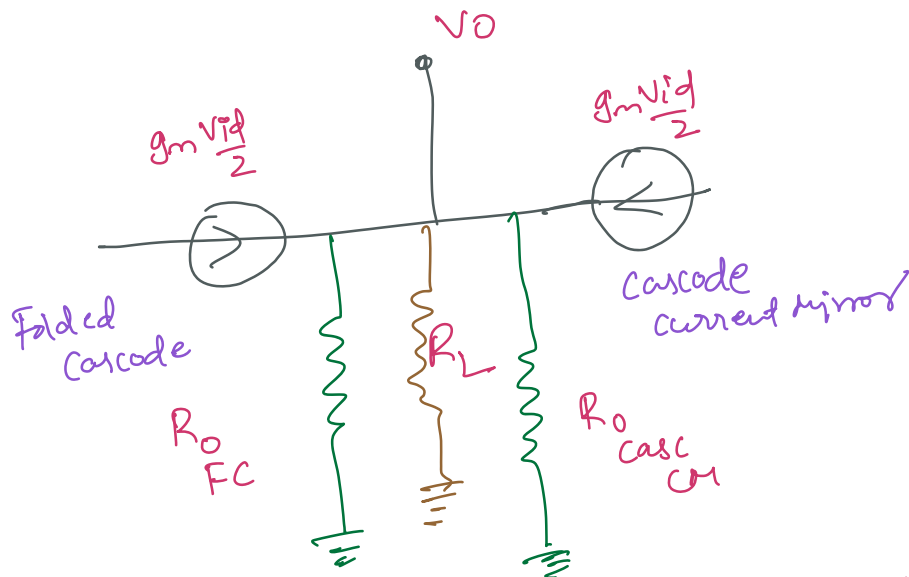
loading of this model is implemented
using Current Mirror.

Highest conductance

→ path ② has high conductance because it is the input conductance of a Common Gate stage

$$G_{icg} \gg \gg \gg g_o \rightarrow \text{conductance of MOS.}$$

→ Currents through paths ③ & ④ is almost negligible



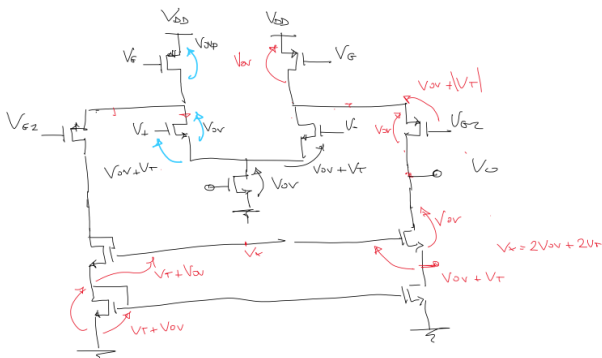
$$\frac{V_O}{v_{id}} = g_m \left(R_{out_{FC}} \parallel R_{out_{casc CM}} \right)$$

Note:- We should make sure that the loading resistance that we connect to the output is sufficiently larger (or equivalent to the $R_{o_{FC}} \parallel R_{o_{casc_{CM}}}$)

As R_L will be parallel to both

$$\frac{V_o}{V_{id}} = g_m \left(R_{out_{FC}} \parallel R_{out_{cas}} \parallel R_L \right)$$

→ Biasing analysis of The above circuit



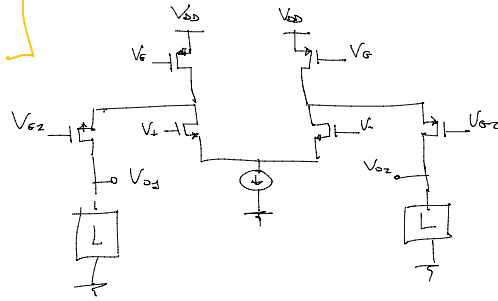
$$V_{O \min} = 2V_{OV} + V_T$$

$$V_{D_{MAX}} = V_{DD} - 2V_{OV} \quad (V_{G2} = V_{DD} - 2V_{OV} - |V_T|)$$

$$\boxed{V_{IN_{CM\ MIN}} = 2V_{OV} + V_{TN}}$$

$$[V_{INCM \text{ MAX}} = V_{DD} - V_{OV_P} - V_{OV_N} + V_T]$$

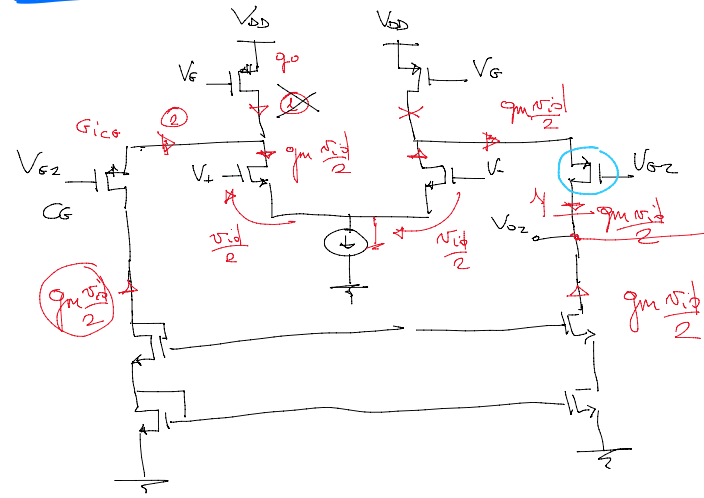
[Folded Cascode
Diff. Output]



Folded Cascode
Single-ended Output
?

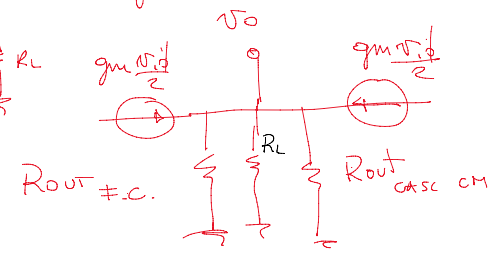
Intuitive Smallsignal Analysis

Folded cascode diff amp / single ended output



$I \rightarrow G \uparrow$

$G_{icg} \gg g_o$



$R_{out} \neq \infty$

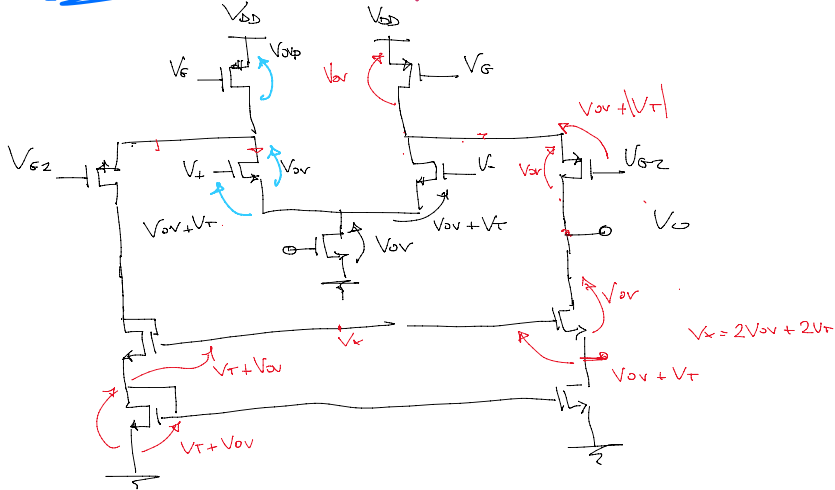
$n // n / D_1$

$$\underline{v_o} = g_m \cdot (R_{out_{tc}} // R_{out_{casc\ C.H.}} // R_L)$$

$$R_1 // R_2 \angle R_1$$

$$\hookrightarrow R_1 < R_2$$

Large Signal Analysis



$$\frac{v_o}{v_{id}} = g_m \cdot (R_{out_{TC}} \parallel R_{out_{CMSC \text{ C.H.}}} \parallel R_L)$$

very Important

Swing of output voltage

$$V_{O \min} = 2V_{OV} + V_T$$

$$V_{O_{MAX}} > V_{DD} - 2V_{OV}$$

$$(V_{G2} = V_{DD} - 2V_{ov} - |V_T|)$$

Swing of Input voltage

$$V_{INCM \text{ MIN}} = 2V_{OV} + V_{TN}$$

$$\left[V_{INCM \text{ MAX}} = V_{DD} - V_{OV_P} - \cancel{V_{OV_N}} + \cancel{V_{OV_N}} + V_T \right]$$

NOTE:-

$$V_{\text{max}}$$

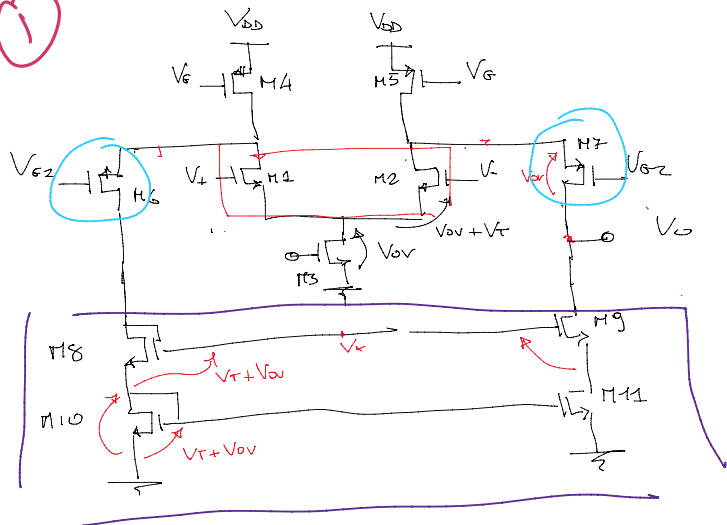
and

Ninain
Car

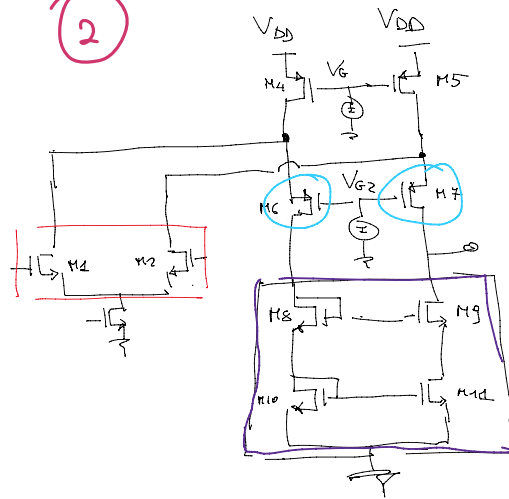
do not have any relation in this

Case.

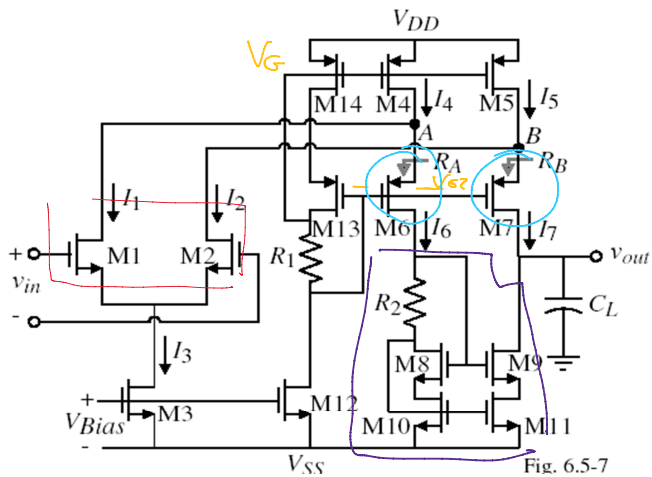
①



②

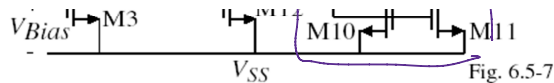


③



① = ② = ③

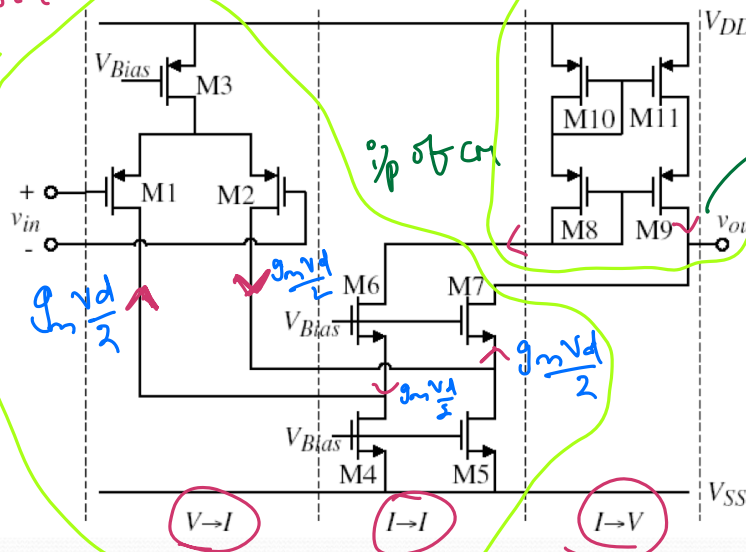
Fig. 6.5-7



CMOS opAmp

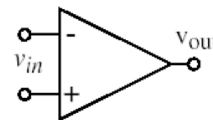
CMOS OpAmp

PMOS Folded Cascode
Diff pair

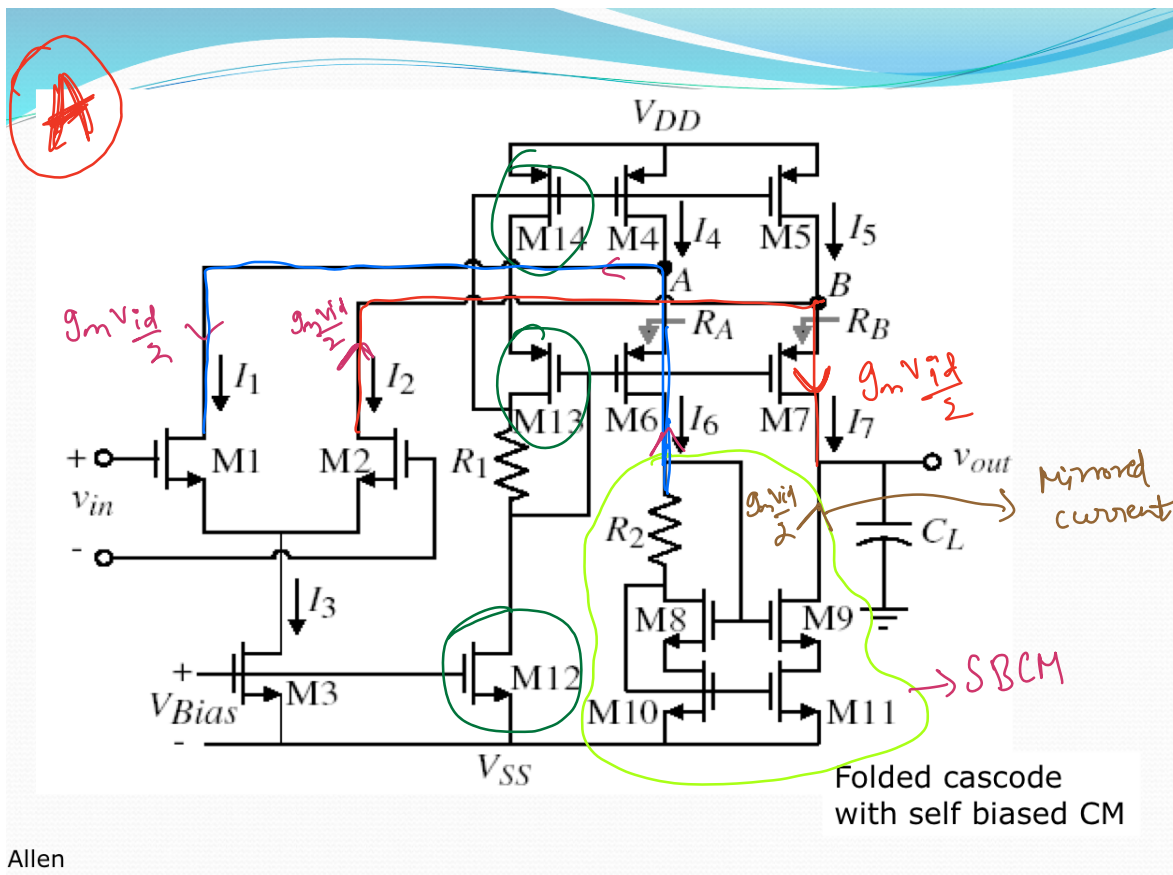


Current Mirror
using PMOS

Mirrored current



✱✱

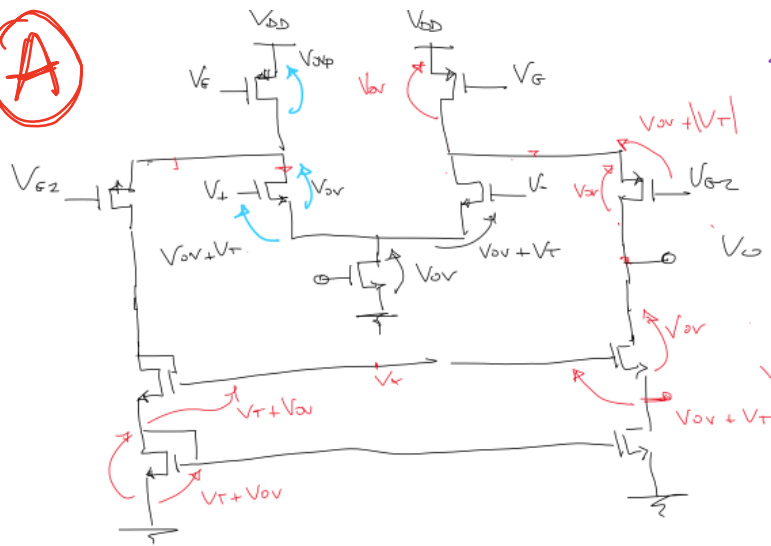


→ M_6, M_7 common Gate stage Transistors

M_8, M_9, M_{10}, M_{11} Form a Self Biased Current mirror

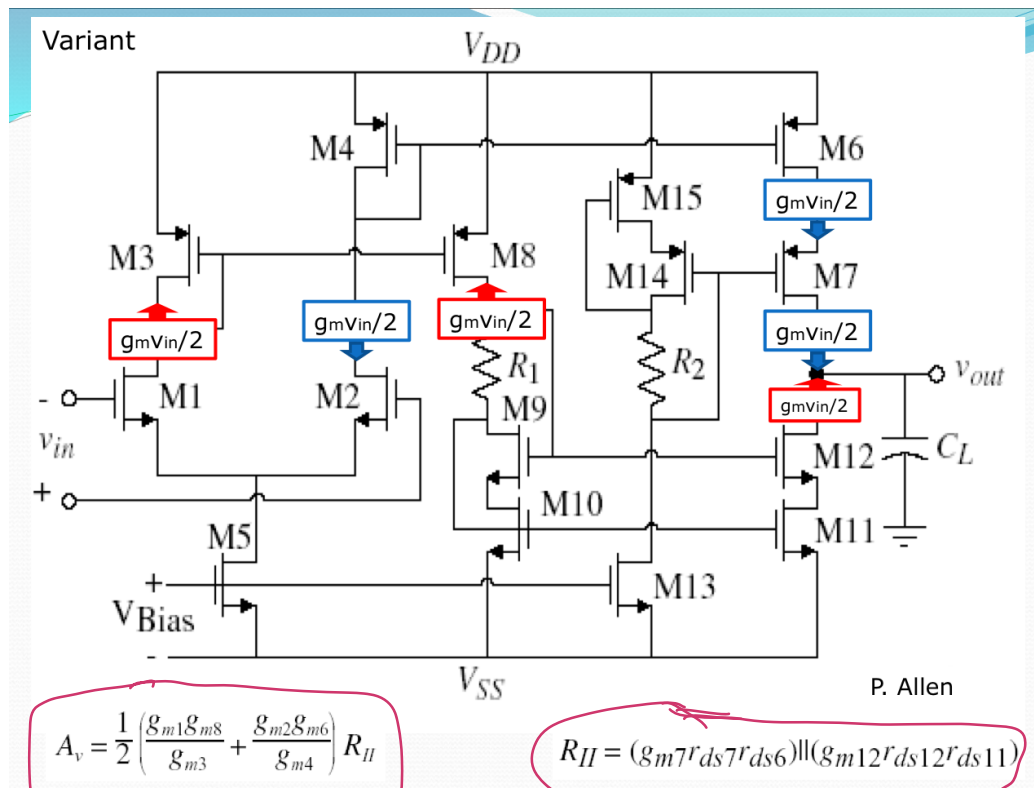
→ M_{12}, M_{13}, M_{14} are here to provide constant Gate voltage

(A)



Both circuits are similar except for the current mirror load.

* Another variant of CMOS op-Amp



$$A_v = \frac{1}{2} \left(\frac{g_{m1} g_{m8}}{g_{m3}} + \frac{g_{m2} g_{m6}}{g_{m4}} \right) R_{II}$$

$$R_{II} = (g_{m7} r_{ds7} r_{ds6}) \parallel (g_{m12} r_{ds12} r_{ds11})$$

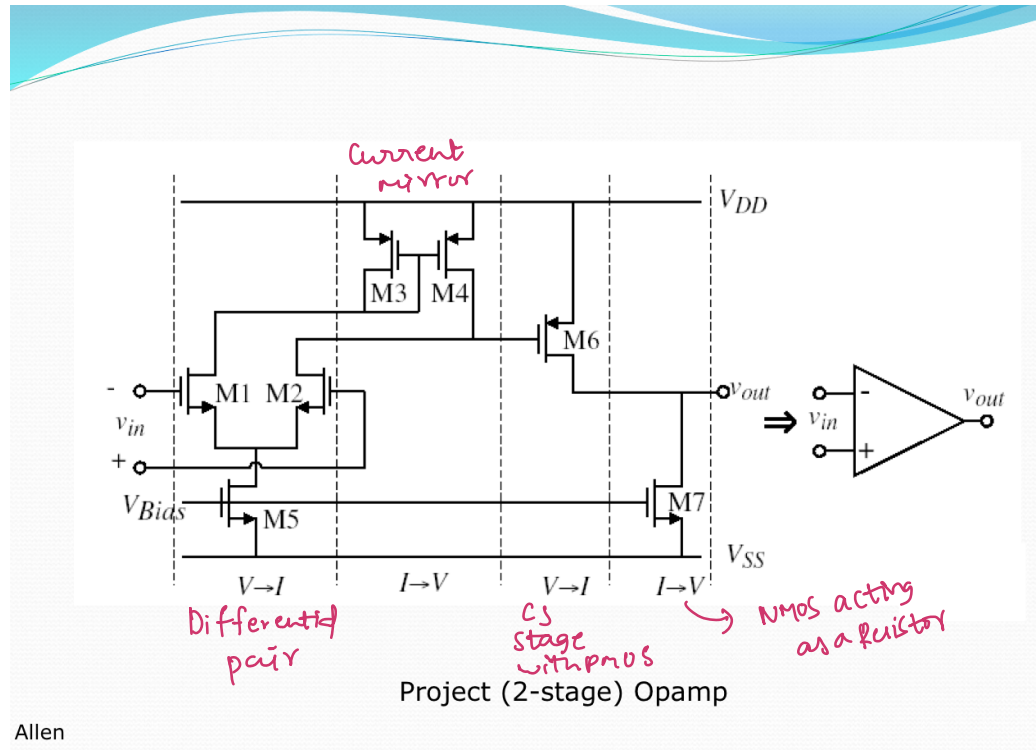
Note If you see we have started from a
Single Transistor Amplifier to an 15
Transistors op-Amp.

→ M_4, M_6 are Transistors in Current Mirror.
 M_7 is a Common Gate (Current Buffer)
stage

→ M_3, M_8 form Current Mirror

M_8 is facing a topology constituted by

$R_1, M_9, M_{10}, M_{11}, M_{12}$ altogether form
Self Biased Cascode Current Mirror.



This is the configuration in which
the project should be done

But till now we studied only

(i) DC response

(ii) AC response

But one of the most important aspect
of circuit analysis is Frequency
Response[#]
will

which we study now -

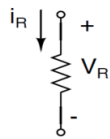


Frequency Response of Analog Integrated Circuits

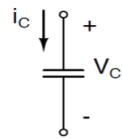


Frequency Response

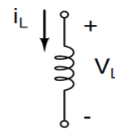
- All linear circuits we deal with can be represented by a rational function of "s" in the Laplace domain.
 - If we are interested in the steady-state response only, Fourier transform will be sufficient.
- Why is frequency domain more convenient?
 - In LTI systems, the response to an arbitrary input can be found through convolution the impulse response.
 - Convolution in time domain transforms to simple multiplication in frequency domain



Resistor



Capacitor



Inductor

	Resistor	Capacitor	Inductor
Time Domain	$V_R(t) = R \times I_R(t)$	$I_C(t) = C \frac{dV_C}{dt}$	$V_L(t) = L \frac{di_L}{dt}$
Frequency Domain	$V_R(s) = R \times I_R(s)$	$I_C(s) = sC V_C(s)$	$V_L(s) = sL I_L(s)$
Impedance	$Z_R = R$	$Z_C = \frac{1}{sC}$	$Z_L = sL$

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Poles & Zeros

- The transfer function of a given circuit can be written as:

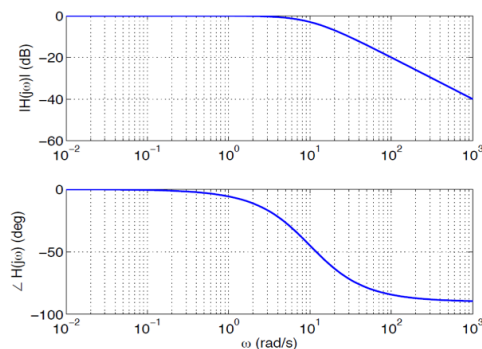
$$TF(s) = \frac{a_0 + a_1s + a_2s^2 + \dots + a_ns^n}{b_0 + b_1s + b_2s^2 + \dots + b_ms^m}$$

- Zeros** are roots of the numerator.
- Poles** are roots of the denominator.
- If we know poles and zeros we can tell how the circuit responds to various frequencies.
 - Crucial in building frequency-selective circuits such as filters
 - Stability of feedback circuits depends on pole and zero locations too (more later)

3



Bode Plot



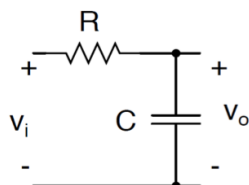
- A Bode plot is composed of two subplots showing *magnitude* and *phase* transfer function.
 - A log-log plot of the magnitude along with a log-angle plot of the phase.
- Evaluate $H(s)$ at $s=j\omega$:
 - Returns the steady-state phasor analysis of the circuit

4



Single-Pole Circuit

- A simple RC low-pass filter has a single pole.



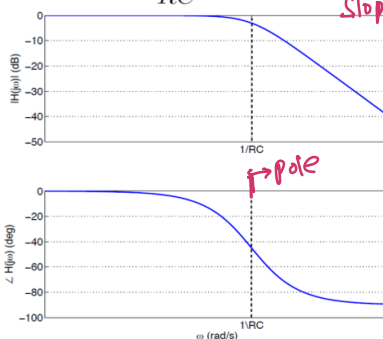
$$H(s) = \frac{v_o}{v_i} = \frac{Z_C}{Z_C + R} = \frac{1}{1 + sRC} = \frac{1}{1 - \frac{s}{p}}$$

$$p = -\frac{1}{RC}$$

$$|H(j\omega)| = \left| \frac{1}{1 + j\omega RC} \right| = \frac{1}{\sqrt{1 + (\omega RC)^2}}$$

$$\angle H(j\omega) = \tan^{-1}(-\omega RC)$$

- At $\omega = 1/RC$ the magnitude drops by 3dB compared to its maximum value.
- At $\omega = 1/RC$ the phase is exactly -45° .



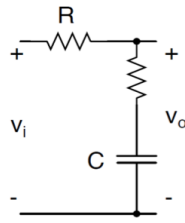
-6 dB
oct

Slope = -20 dB
dec

→ The objective of the pole is to generate loss in the amplitude of the transfer function & 90° loss in phase response of this circuit.

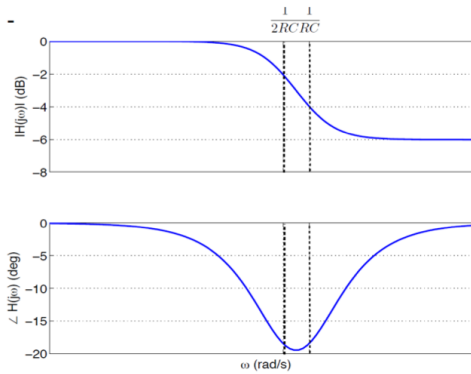


A Circuit w Pole and Zero



$$H(s) = \frac{v_o}{v_i} = \frac{R + Z_C}{Z_C + R + R} = \frac{1 + sR}{1 + s2R}$$

$$p = -\frac{1}{2RC}, \quad z = -\frac{1}{RC}$$



6

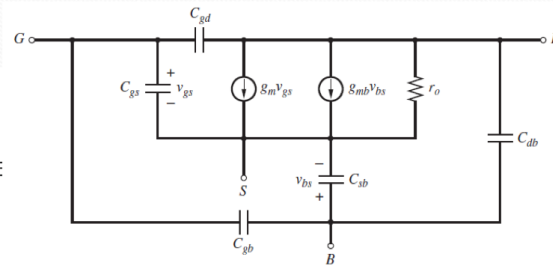
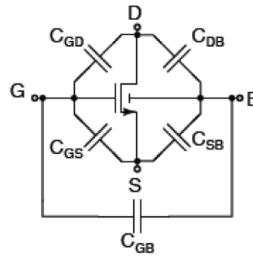
positive zero acts as -1 for Amplitude
& phase loss for phase

negative zero phase gain for phase

$+1$ for Amplitude



MOSFET: Capacitance Summary

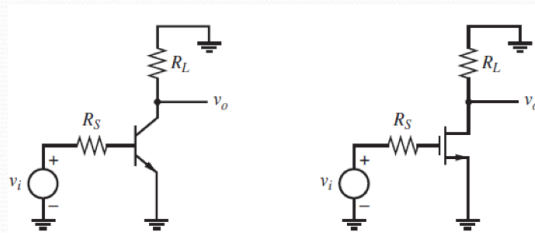


	Cut-off	Triode	Active
C_{GS}	C_{OV}	$\frac{1}{2}WLC_{ox} + C_{OV}$	$\frac{2}{3}WLC_{ox} + C_{OV}$
C_{GD}	C_{OV}	$\frac{1}{2}WLC_{ox} + C_{OV}$	C_{OV}
C_{GB}	$\left[\frac{1}{C_{CB}} + \frac{1}{WLC_{ox}}\right]^{-1}$	0	0
C_{SB}	C_{jdb}	$C_{jdb} + \frac{C_{CB}}{2}$	$C_{jdb} + \frac{2}{3}C_{CB}$
C_{DB}	C_{jdb}	$C_{jdb} + \frac{C_{CB}}{2}$	C_{jdb}

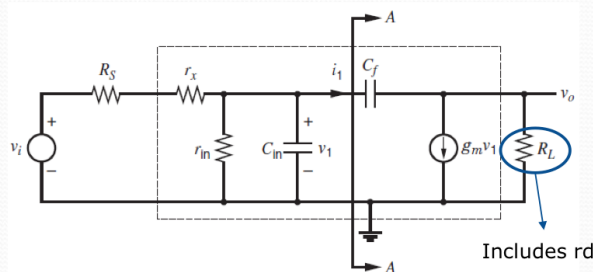
unbalance b/c C_{db} & C_{gb}

7

Frequency Response of a Common Source (Emitter) Amplifier



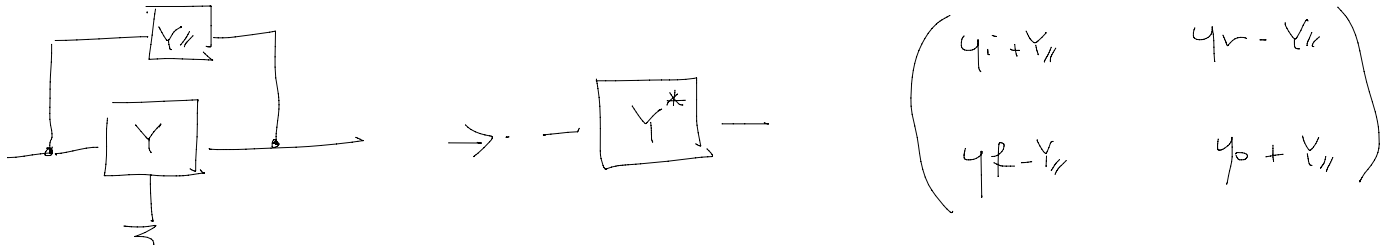
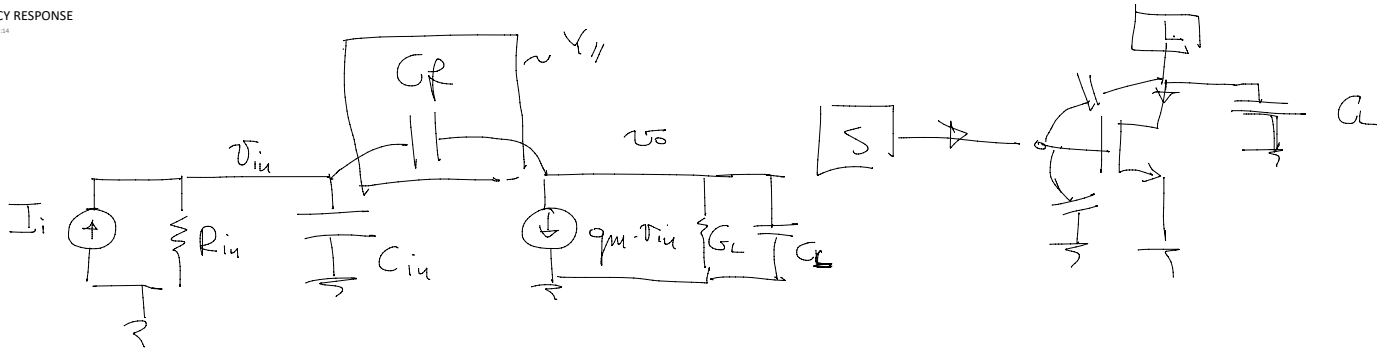
General Model	Bipolar Model	MOS Model
r_x	r_b	0
r_{in}	r_π	∞
C_{in}	C_π	C_{gs}
C_f	C_μ	C_{gd}
r_o	r_o	r_o



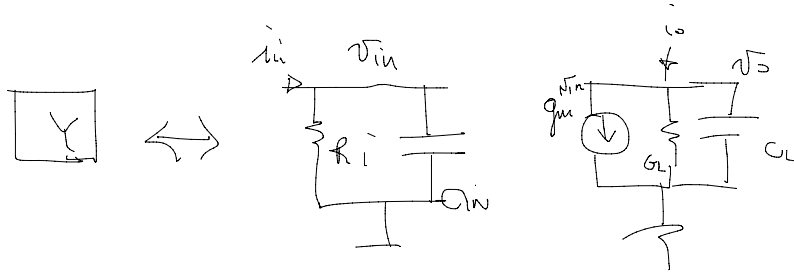
$$R = (R_S + r_x) || r_{in}$$

$$i_i = \frac{v_i}{R_S + r_x}$$

Norton equivalent



$$y_i = \left. \frac{i_i}{v_i} \right|_{v_o=0} \quad y_f = \left. \frac{i_o}{v_i} \right|_{v_o=0} \quad y_o = \left. \frac{i_o}{v_o} \right|_{v_i=0} \quad y_r = \left. \frac{i_i}{v_o} \right|_{v_i=0}$$



$$G_i = \frac{1}{R_i}$$

$$Y_{//} = sC_f$$

$$Y = \begin{pmatrix} G_{in} + sC_{in} & - \\ g_m & G_L + sC_L \end{pmatrix} \quad Y^* = \begin{pmatrix} G_{in} + sC_{in} + sC_f & -sC_f \\ \underline{g_m - sC_f} & G_L + sC_L + sC_f \end{pmatrix}$$

$$\left[\frac{v_o}{i_i} = - \frac{y_f^*}{\Delta Y^*} \right]$$

$$\left[\frac{v_o}{v_i} = - \frac{y_f^*}{\Delta Y^*} \cdot G_{in} \right]$$

$$A_v = \frac{-(g_m - sC_f) \cdot G_{in}}{[G_{in} + s(C_{in} + C_f)][G_L + s(C_L + C_f)] + sC_f(g_m - sC_f)}$$

$$\left[s_z = \frac{g_m}{C_f} \right]$$

$$\left(1 - \frac{s}{s_z} \right)$$

$$s_p = \frac{C_f + C_L}{C_{in} + C_f} + \frac{g_m C_f}{C_{in} + C_f}$$

$$= -\frac{g_m}{G_{IN}} \cdot \frac{\left(1 - \frac{s}{s_z}\right)}{\left(1 + as + bs^2\right)}$$

$$a = \frac{C_f + C_L}{G_L} + \frac{C_{IN} + C_f}{G_{IN}} + \frac{g_m C_f}{G_L \cdot G_{IN}}$$

$$b = \frac{C_{IN} \cdot C_f + C_{IN} C_L + C_f \cdot C_L}{G_{IN} \cdot G_L}$$

$$\begin{aligned} & \left(1 - \frac{s}{p_1}\right) \left(1 - \frac{s}{p_2}\right) = \\ & = 1 - \frac{s}{p_1} - \frac{s}{p_2} + \frac{s^2}{p_1 p_2} \approx 1 - \frac{s}{p_1} + \frac{s^2}{p_1 p_2} \end{aligned}$$

$$-\frac{1}{p_1} \approx a \rightarrow p_1 \approx -\frac{1}{a}$$

$$b = \frac{1}{p_1 p_2} \rightarrow p_2 \approx -\frac{a}{b}$$

$$\textcircled{p_2 \gg p_1}$$

Gain CS

$$p_1 \approx -\frac{G_{IN} \cdot G_L}{g_m \cdot C_f}$$

$$a \approx \frac{g_m \cdot C_f}{G_L \cdot G_{IN}}$$

T.C.

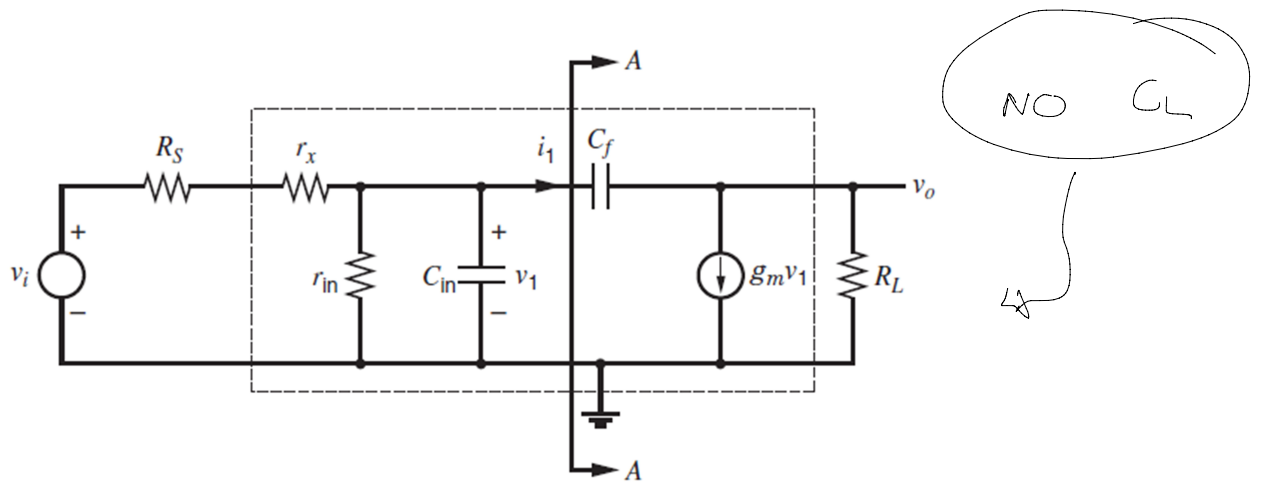
$$= \left(\frac{g_m}{G_L} \cdot C_f \right) \cdot R_{IN}$$

$$p_2 = -\frac{\frac{g_m \cdot C_f}{G_L \cdot G_{IN}}}{\frac{C_{IN} \cdot C_f + C_{IN} C_L + C_f \cdot C_L}{G_{IN} \cdot G_L}}$$

$$= -\frac{g_m \cdot C_f}{C_{IN} \cdot C_f + C_{IN} \cdot C_L + C_f \cdot C_L}$$

$$\text{if } C_L = 0 \rightarrow p_2 \approx -\frac{g_m}{C_{IN}}$$

$$C_L \uparrow \quad C_f \gg C_{IN} \quad p_2 \approx -\frac{g_m}{C_L} \sim g_m \cdot C_{II}$$



$$p_1 = -\frac{1}{C_{in}R + C_f(R + g_m R_L R + R_L)} = -\frac{1}{R \left[C_{in} + C_f \left(1 + g_m R_L + \frac{R_L}{R} \right) \right]}$$

$$p_2 = \frac{1}{p_1} \frac{1}{R_L R C_f C_{in}} = -\left(\frac{1}{R_L C_f} + \frac{1}{R C_{in}} + \frac{1}{R_L C_{in}} + \frac{g_m}{C_{in}} \right)$$

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